

N that pays, where it pays, and where to look next

An orchestra of R packages that refuses the crop model where the field disagrees, prices the decision, and says what the next test is worth

Max Moldovan · Adelaide University · ORCID 0000-0001-9680-8474 max578.github.io/orchestra

The problem

A crop model gives a nitrogen recommendation at every paddock, including the paddocks where it is wrong. A field trial says what happened, not why. Neither, alone, can tell a grower **where** to trust the model, **what** to do where it fails, and **which** paddock is worth one more soil test or strip trial.

Most analytics pipelines answer anyway. The design decision here is that every component may **refuse**, the refusal is a typed object every downstream component reads, and no recommendation is minted from a posterior the field has rejected.

The orchestra

A federation of open R packages typed by one S7 `orchestra_manifest` contract and one predicate, `is_orchestra_decline()`. The chain used here:

apsimR APSIM Next Gen as a member – the process-based truth
terroir free climate and soil at any point (NASA POWER, SoilGrids)
PESTO simulator inversion – iterative ensemble smoother, real PEST++
flexyBayesOrchestra surrogate emulators so inversion never loops the simulator
kernR TACI – the theory-anchored causal test: does the field agree?
brms lens the field record read through plateau-capable curves
decideR loss, expected value of perfect and sample information
optimix which sites to test under a budget
grainPlan the grower-facing plan, which also refuses
masque sovereign synthetic twins of real trial data

16 members and 7 candidates; every member's refusal passes one fleet gate (20 of 20, 2026-08-26).

Study A0 -- the set-up

A **Synthetic Barley Belt**: 40 sites on real grain-belt coordinates with real climate and soil, 5 seasons, 6 N rates x 3 varieties x 3 replicates, yields from APSIM.

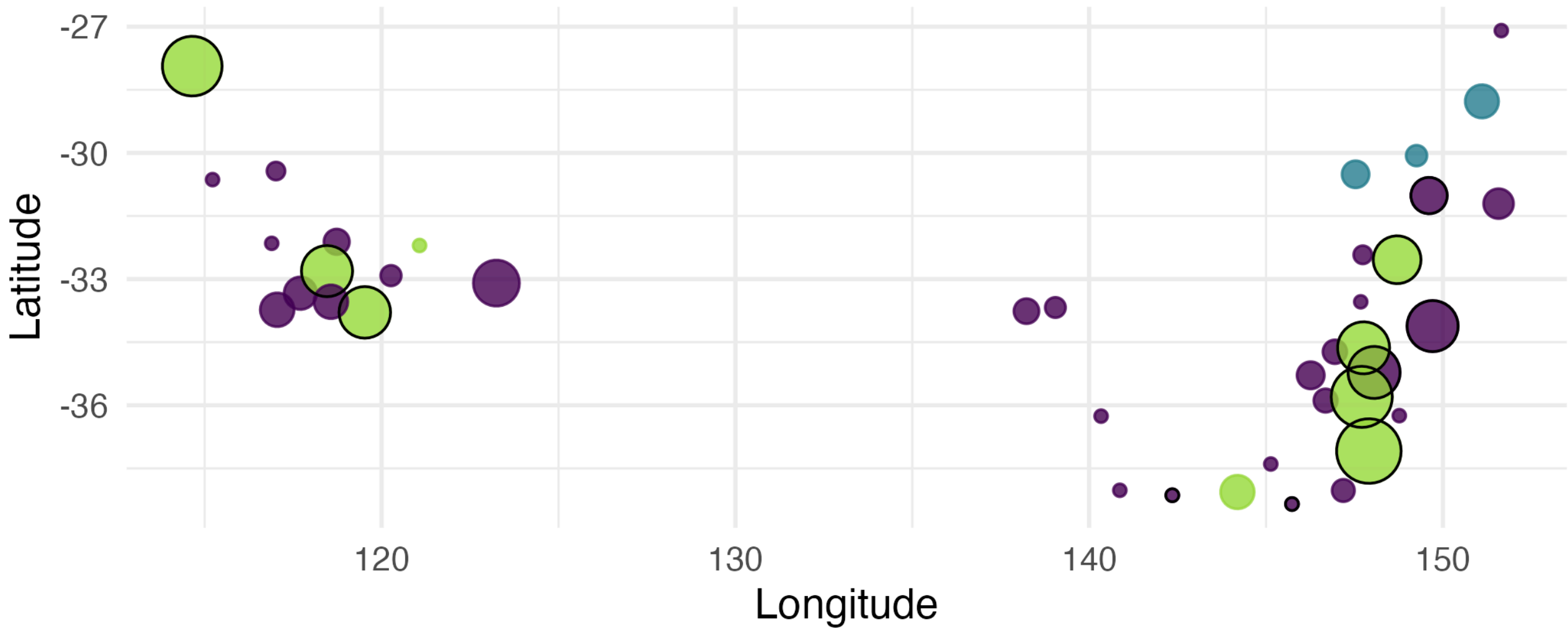
At 12 sites a **hidden subsoil constraint** caps yield and flattens the N response above 60 kg N/ha. The simulator used for inversion is never told.

The question at every site-season: which rate maximises profit net of nitrogen and its nitrous-oxide cost, where should the model be refused, and what is one more measurement worth?

Every stage carries an independent oracle inside its script; the report binds every number to an object; the whole chain is re-verified on a re-seeded belt.

A Where the mechanism was refused and the field record took over

40 sites on real grain-belt coordinates (NASA POWER climate, SoilGrids soil); black ring = hidden subsoil constraint planted (12 sites)

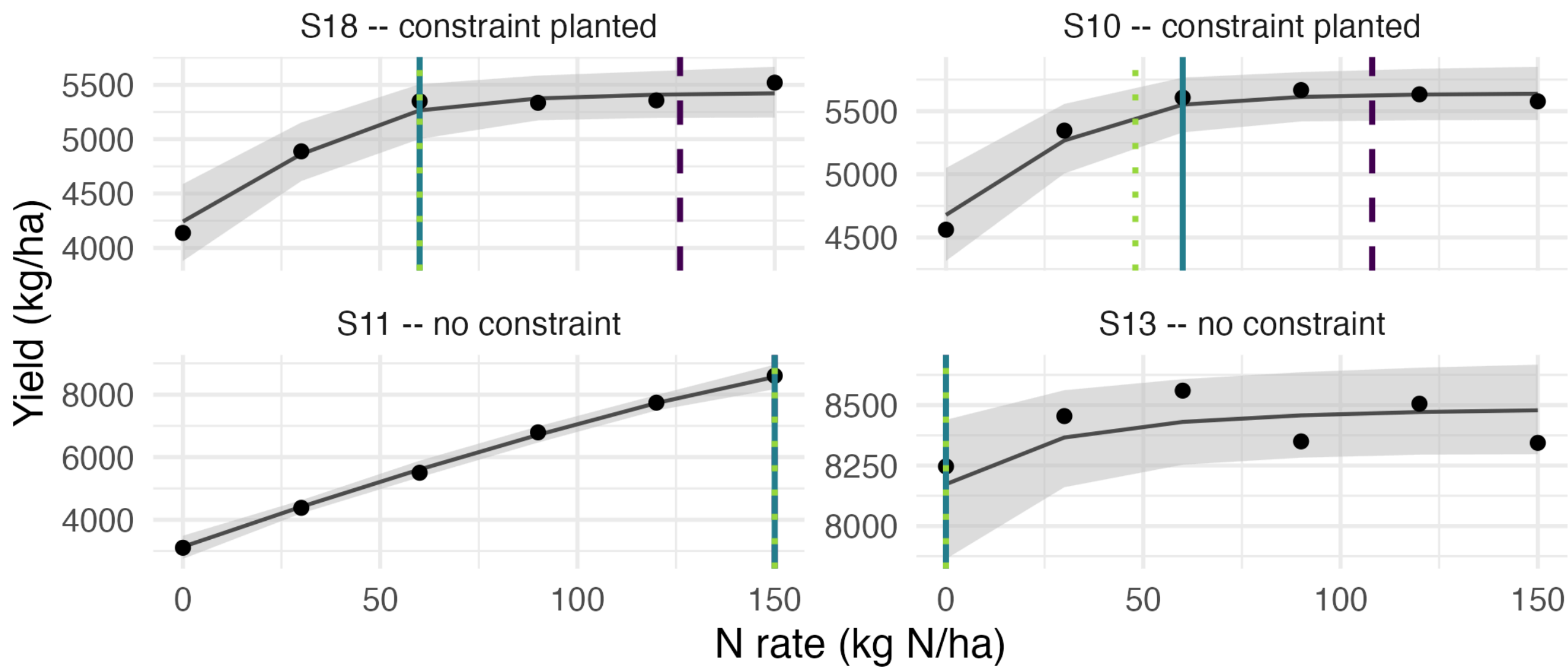


Lens the cascade used ● mechanism ● mixed ● statistical

Naive loss (AUD/ha, season mean) ● 0 ● 50 ● 100

B What each lens recommends at four sites

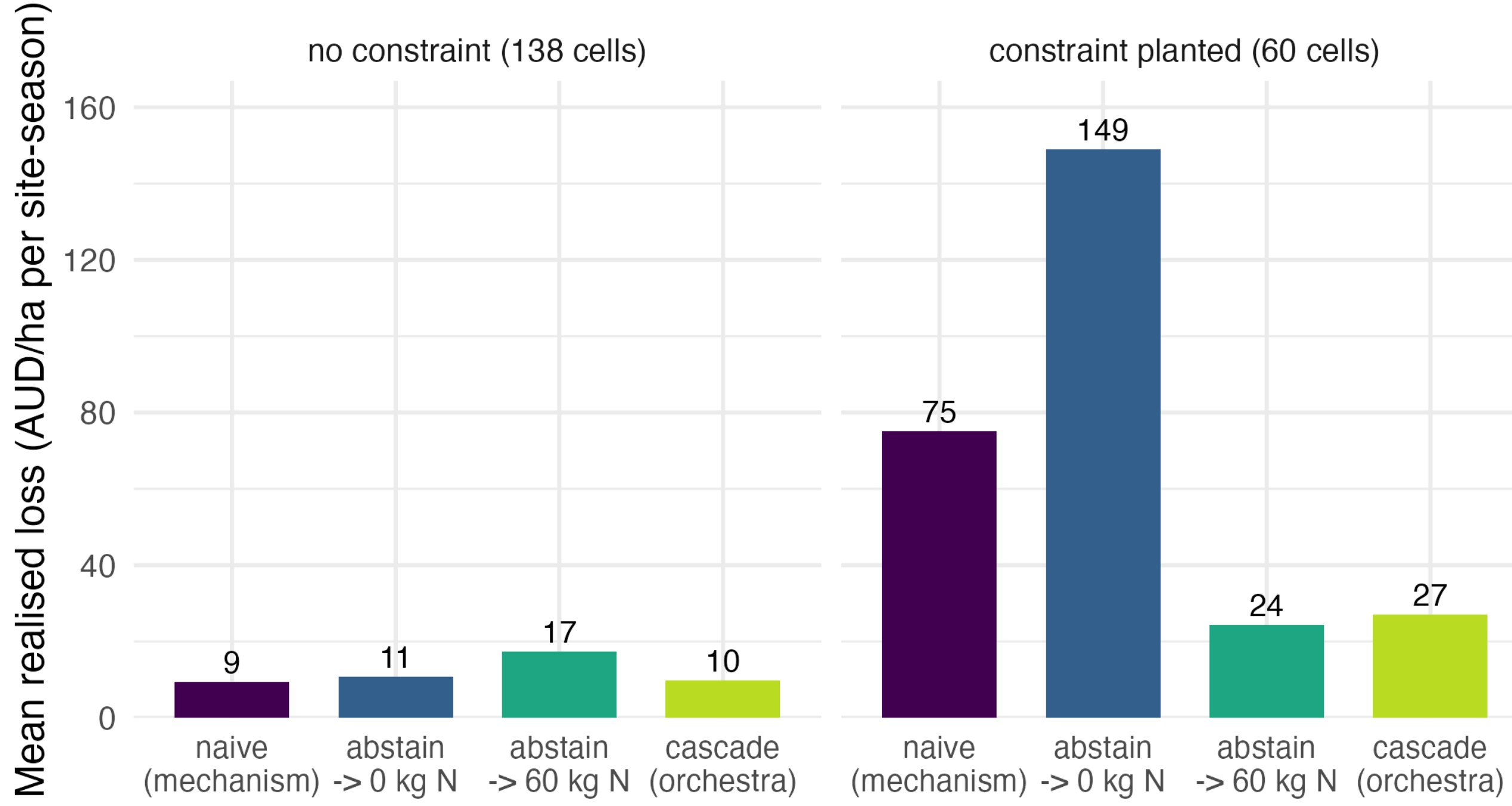
Grey: the field-record lens (plateau mixture, posterior mean and 90% band); points: observed plot means; lines: season-mean N rate



Season-mean N rate | naive (mechanism only) | cascade | truth optimum

C What each policy loses against the planted truth

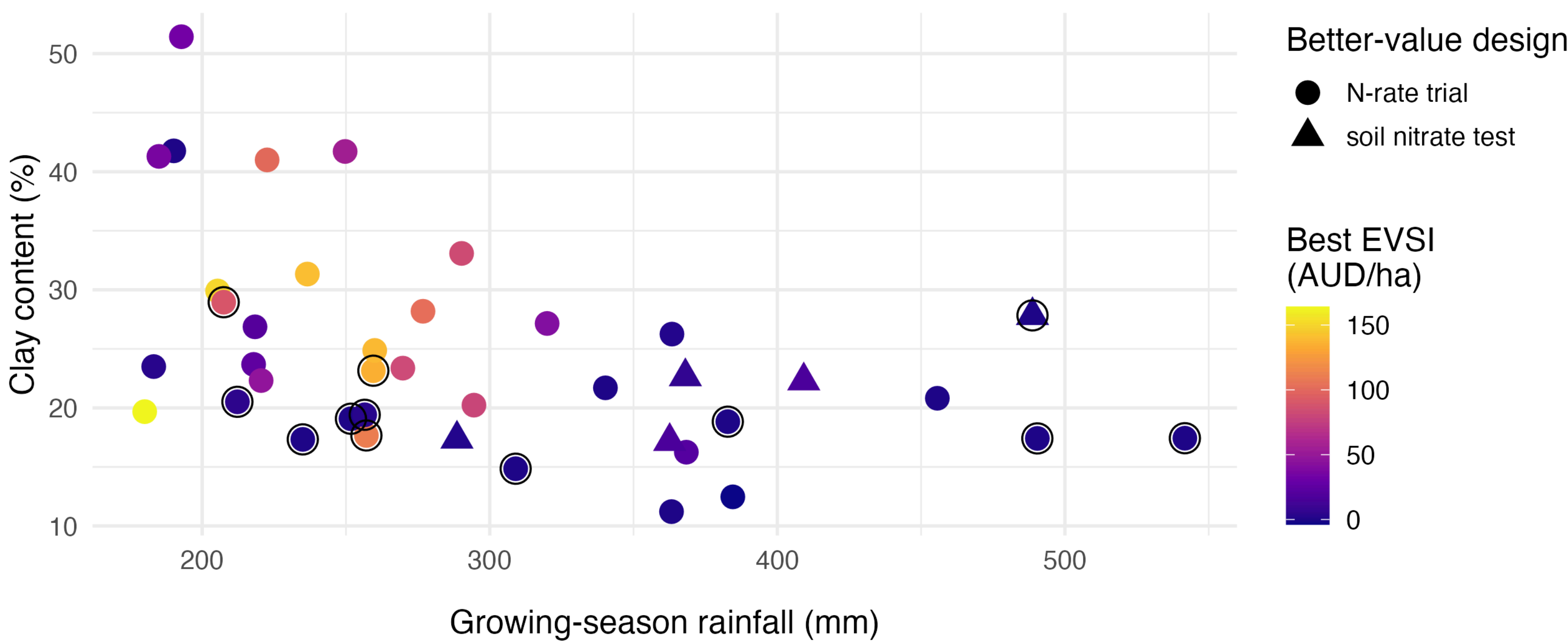
Belt total: cascade 2,960 vs naive 5,818 AUD/ha summed over 198 cells (49% less loss); re-seeded belt 2,549 vs 5,002



Grain price 314.5 AUD/t, nitrogen 2.07 AUD/kg N incl. N2O shadow price; information prices [unverified]. No GRDC data. Every number binds to studies/A0/output/poster_numbers.csv.

D Where the next research dollar pays: expected value of sample information

Top site S33 at 164 AUD/ha; 40 of 40 sites priced; black ring = constraint planted; directional (importance-weighted EVSI, REPORT.md section 9)



Three numbers

2,960 vs 5,818 AUD/ha

Summed loss against the planted truth, orchestra cascade vs the naive mechanism-only arm: **49 per cent less loss**. At the constrained cells 27 vs 75.2 AUD/ha per site-season; at the unconstrained cells 9.7 vs 9.5. Reproduced on a re-seeded belt (2,549 vs 5,002).

1,440 kg N not applied, 6.2 t CO2-e/ha avoided

The field record locates the plateau the model cannot see: the posterior interval of the plateau onset contains the planted 60 kg N/ha at 12 of 12 constrained sites (median onset 67 vs 97 kg N/ha elsewhere). TACI refuses the mechanism with sensitivity 0.58 and specificity 0.93.

164 AUD/ha top-site value of one strip trial

40 of 40 sites priced for the expected value of sample information – the map of where the next research dollar pays. Directional, not calibrated (see right).

What it does not show

- A synthetic belt, not a real one: the shape conclusions have real-data cover (the belt's N-response slopes partially bracket a real barley multi-environment trial), the dollar magnitudes do not.
- Two of seven stage oracles fail and are reported as failures: the plateau-onset separation between constrained and unconstrained sites (AUC 0.86 against a stated 0.9 – unconstrained sites plateau too), and an under-powered site-level test on the re-seeded belt.
- EVSI is importance-weighted and effective-sample-size limited at this posterior size; information prices are uncalibrated and await validation.
- Economics: barley 314.5 AUD/t, nitrogen 2.07 AUD/kg N plus an IPCC-default N2O shadow price.
- No GRDC data enters the belt, the figures or this poster.

Try it



max578.github.io/orchestra

Member atlas, the study series, the contract, and installation from `max578.r-universe.dev`. Study A0 runs from `studies/A0/run_A0.R`; the five-minute offline walk is `demo/a0_demo.R`.

Where next. B0: climate-forward genomic selection. C1: impact attribution for programme investment. On A0: a conjugate fast path for EVSI, and the plateau lens as a package verb.